

Test 1 · Algebra

Grade 8

40 minutes 40 marks 21 questions No calculator

1–3 Revision of the Grade 7 course

4–5 Algebraic expressions

6–8 Algebraic fraction. Cancelling fractions

Algebra 8, p. 3 · Stage 8 review

Algebra 8, §1 · Stage 9 · 2.1–2.2

Algebra 8, §2 · Stage 9 · 2.5

- Answer every question. Show your working — method carries marks.
- Whenever you cancel a fraction, state the permissible values of the variable.
- No calculator.

NAME _____ CLASS _____ DATE _____ MARK _____ / 40

A Section A · Short answers

EASY 10 marks

One mark each. A single line of working is enough.

1 Simplify: $x^6 \cdot x^4$

2 Simplify: $(b^4)^3$

3 Expand: $4(3x - 2)$

4 Factorise: $10a - 15$

5 Which two of these are fractional expressions?

$$\frac{2x}{7} \quad \frac{4}{x} \quad \frac{x}{3} - 5 \quad \frac{9}{x+2}$$

6 State the permissible values of $\frac{1}{x-6}$

7 Find the value of $\frac{x+5}{x-2}$ when $x = 3$

8 Cancel: $\frac{8a}{12a}$

9 Cancel: $\frac{15x^3}{5x}$

10 Cancel: $\frac{5(x-2)}{10(x-2)}$

B Section B · Two marks each

MEDIUM 14 marks

Method marks are available even when the final answer is wrong.

11 Simplify: $(3a^3b^2)^2$ [2]

12 Expand and simplify: $(x+5)(x-2) - x^2$ [2]

13 Factorise: $9x^2 - 16y^2$ [2]

14 Cancel: $\frac{3x + 12}{x^2 - 16}$ [2]

15 State the permissible values of $\frac{x + 2}{x^2 - 25}$ [2]

16 Find the value of $\frac{a^2 - b^2}{a + b}$ when $a = 9$, $b = 4$ [2]

17 For which value of x is $\frac{x - 7}{x + 1}$ equal to zero? [2]

C Section C · Four marks each

HARD

16 marks

Factorise before you cancel, and state the values the variable may not take.

18 (a) Factorise completely: $x^3 - 9x$ [2]

(b) Hence cancel $\frac{x^3 - 9x}{x^2 + 3x}$ and state the permissible values. [2]

19 Cancel $\frac{x^2 - 8x + 16}{x^2 - 16}$, state the permissible values, then find the value of your answer when $x = 9$.
[4]

20 (a) Cancel: $\frac{2x^2 - 2y^2}{3y - 3x}$ [3]

(b) Explain in one sentence why $\frac{x + 5}{5}$ cannot be cancelled. [1]

21 Cancel $\frac{2x^2 + 5x - 3}{x^2 - 9}$, stating the permissible values. [4]